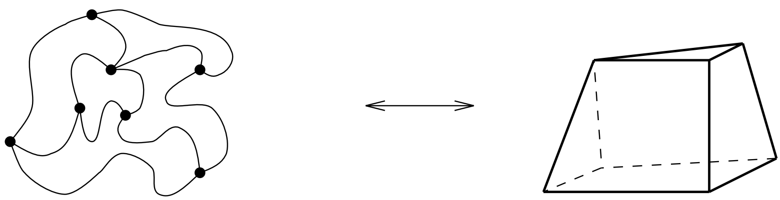
4
Steinitz' Theorem for 3-Polytopes

The combinatorial structure of 2-polytopes is not much of a mystery. For an illustrated journey into the wonderful world of 3-dimensional convex (and nonconvex) polyhedra, we direct the reader's attention to the conference volume [492]. See also Barnette's book [45] for a nice elementary discussion of combinatorics and graph theory related to convex 3-polytopes.

In this lecture we will establish the basic theory for 3-polytopes, by proving *Steinitz' theorem*. The basic version reads as follows.

Theorem 4.1 (Steinitz' theorem). [524, 527]
G is the graph of a 3-dimensional polytope if and only if it is simple, planar, and 3-connected.

Polytopal graphs are certainly *simple*: they have no loops or multiple edges. The graph $G(P)$ is planar for every 3-polytope (use radial projection to a sphere from an interior point, or a linear projection to a plane from a point beyond a facet). Also, $G(P)$ is 3-connected by Balinski's Theorem 3.14. Thus, the difficult part is the "if" part of the theorem: it requires that we show how, given a 3-connected planar graph, one can construct a 3-polytope.



DOI 10.1007/978-1-4613-8431-1_5, © Springer Science+Business Media New York 2007

Here are four observations to indicate to this claim is *nontrivial*:

1. No similar theorem is known, and it seems that no similarly effective theorem is possible, in higher dimensions.

2. There are lots of interesting consequences and various strengthenings that follow by the same proof technique (see Section 4.4).

3. Many of the higher-dimensional analogues of these strengthenings are false (we'll see this in Lectures 5 and 6).

4. There is no extremely simple proof known.

All the combinatorial ("classical") proofs of Steinitz' theorem [524] [527, §§54,63] [252, Sect. 13.1] [51] basically follow the same pattern: arguing that every 3-connected planar graph can be "built up" from K_4 by some well-defined operations, which preserve realizability. (See the notes at the end of this lecture for a different, "nonlinear" line of reasoning.)

The reason why we can give a "nicer than usual" combinatorial proof here is that we will be more elegant in dealing with the graph theory: using Truemper's clever treatment [546, 547] of "ΔY reductions." This will be done in the next three sections of this lecture, which together imply Theorem 4.1.

– In Section 4.1, we discuss the little graph theory we need, concentrating on 3-connected graphs and ΔY reductions.

– In Section 4.2, we prove that Steinitz' theorem is true for 3-connected planar graphs which have a "ΔY reduction."

– In Section 4.3, we show that if a 3-connected planar graph G has a ΔY reduction, then so does every minor of G.

– Finally, we show that every planar graph is a minor of a grid graph, and every grid graph has a ΔY reduction.

We close the lecture in Section 4.4 with a list of strengthenings, extensions, and corollaries of Steinitz' theorem.

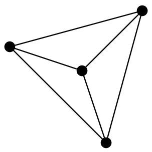
4.1 3-Connected Planar Graphs

Again we need some basic graph theory. Let us, for a while, admit nonsimple graphs as well, that is, graphs that can have loops and parallel edges.

Such a graph G is *connected* if there is a path between any two distinct vertices of G. All the graphs we consider will be connected.

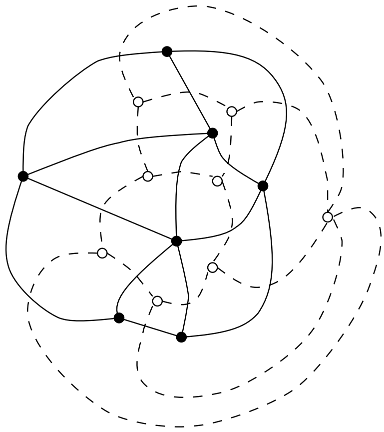
A graph G with at least 2 edges is 2-*connected* if it is connected, has no loops, and cannot be disconnected by removing one vertex and all the edges incident with it. A graph G with at least 4 edges is 3-*connected* if it is simple and cannot be disconnected by removing 1 or 2 vertices from G. Under this definition, the smallest 2-connected graph is the graph C_2 with two parallel edges

and the smallest 3-connected graph is K_4,



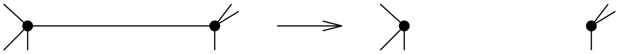
the complete graph with 4 vertices and 6 edges.

The reason for this version of the definitions is that it is invariant under duality. That is, if we embed a graph into the sphere S^2 and draw the *dual graph* G^*, then G is k-connected if and only if G^* is k-connected (for $k = 2, 3$). We do not review the construction of a dual graph here, but trust that the following picture — which you may interpret as being drawn in the plane, or on the 2-sphere — explains it all.

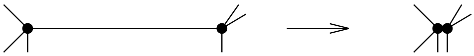


At this point, observe that the combinatorial structure of a 3-polytope is completely determined by its graph — this is a special case of a fact we noted in Remark 3.13(2). It follows from Whitney's theorem [564] that the embedding of a 3-connected planar graph into the sphere is unique. To prove it, note that for every 3-connected planar graph the regions of an embedding are bounded by chordless cycles that do not separate the graph. Any other cycle has more than one region on either of its sides, so either the cycle has a chord, or it separates two vertices, or both. Note that Perles' question of Problem 3.13* asks for a higher-dimensional version of Whitney's theorem.

Two very basic "local" operations on graphs are the *deletion* of edges

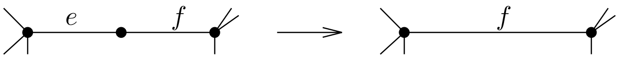


and the *contraction* of edges,

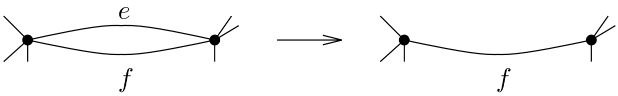


for which the two vertices of the edge are identified. Any graph that can be obtained from G by a sequence of deletions and contractions of edges is called a *minor* of G. Note that the edges of a minor can be viewed as a subset of the edge set of G.

A special case occurs if we contract only edges that are *in series* with others, being adjacent to a vertex of degree 2 (this is equivalent to "removing a subdivision point")

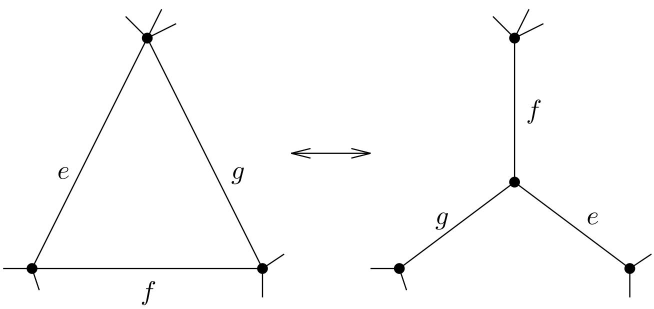


or delete edges that are in parallel with others (this is the usual operation for "making a graph simple").



We will refer to any sequence of such operations as *series-parallel reductions*, or *SP reductions.*

A *Delta-Wye operation*, or *ΔY operation*, replaces a triangle that bounds a face (i.e., a nonseparating triangle) by a 3-star that connects the same vertices, or vice versa. If we want to specify the direction of the transformation, then we will call it a *Δ-to-Y transformation*, respectively a *Y-to-Δ transformation.*



We refer to the figure for the "natural" correspondence between the edges of the triangle and the edges of the 3-star. Note that these operations, replacing a K_3 by a $K_{1,3}$, preserve the number of edges in the graph.

However, a ΔY transformation might create series or parallel edges, which can then be SP-reduced.

We note here a simple lemma, characterizing when connectivity is preserved under Y-to-Δ operations.

Lemma 4.2.

(i) *Let G be a 2-connected graph, and let $\{e, f, g\}$ be the edges at a vertex $\boldsymbol{v}$ of degree 3 in G.*

If none of its edges are parallel (i.e., if $\boldsymbol{v}$ has three different neighbors), then the result of a Y-to-Δ operation is again 2-connected.

(ii) *Let G be a 3-connected graph (in particular, there are no parallel edges; all vertex degrees are at least 3) that is not K_4. Let $\{e, f, g\}$ be the edges at a vertex $\boldsymbol{v}$ in G of degree 3.*

If we perform a Y-to-Δ operation on this 3-star, and then delete all parallel edges created by this (i.e., all edges that originally connected neighbors of $\boldsymbol{v}$), then the resulting graph is again 3-connected.

Proof. This follows directly from the definitions: for this consider a Y-to-Δ operation $G \longrightarrow G'$. Then for any set of one or two separating vertices in G', the same set is separating in G as well. The only problem is that a Y-to-Δ operation can create parallel edges; if we delete them, the operation preserves 3-connectedness. □

The nice thing now is that, because of duality, we immediately get a dual statement, Lemma 4.2*, about connectivity after a Δ-to-Y transformation. For this we use that under duality, we have

$$\begin{array}{rcl}
\text{embedded planar graph } G & \longleftrightarrow & \text{dual graph } G^*, \\
\text{contracting series edges} & \longleftrightarrow & \text{deleting parallel edges}, \\
k\text{-connected} & \longleftrightarrow & k\text{-connected}, \\
\text{nonseparating triangle} & \longleftrightarrow & \text{3-star}, \\
\Delta\text{-to-}Y \text{ transformation} & \longleftrightarrow & Y\text{-to-}\Delta \text{ transformation}.
\end{array}$$

This duality can be carried into our reduction for polytopes, because the graph of a 3-polytope is exactly the dual graph of the polar polytope:

$$G(P)^* \ = \ G(P^\Delta).$$

4.2 Simple ΔY Transformations Preserve Realizability

By a *simple ΔY reduction* we mean any ΔY operation followed immediately by all the SP reductions that are then possible. By Lemma 4.2 and its dual,